IntelliStudy — Design Document

1. Problem Statement

I struggle to keep track of assignments and exams because deadlines are spread across different course platforms. I want to build IntelliStudy, a calendar app that organizes deadlines and study sessions in one place. This would help me manage my time and avoid last-minute cramming.

2. What I'm Going to Build

I'm going to build IntelliStudy, a student planner that lets users add assignments, exams, and study sessions to a calendar. Users will be able to set deadlines, edit tasks, and mark them as completed to keep track of their workload.

3. Eisenhower Matrix

Do — MVP (Important and Urgent)

- Add, edit, and delete assignments, exams, and study sessions with dates.
- View tasks and deadlines on a calendar.
- Mark tasks as completed.
- Save tasks locally so they remain after refreshing.

Schedule — V2.0+ (Important and Not Urgent)

- Send reminders before deadlines.
- Automatically suggest study sessions based on upcoming exams.
- Sync events with Google Calendar.

Delegate — Use a Library (Not Important and Urgent)

- Use a calendar library to display events.
- Use a date picker library for selecting dates and times.
- Use an icon library for edit, delete, and completion buttons.

Delete — Cut It (Not Important and Not Urgent)

- Social feeds and messaging between students.
- Animated backgrounds and elaborate visual effects.
- Points, badges, and competitive leaderboards.

4. Tech Stack

- **Languages:** JavaScript with React for the interface, HTML for structure, and CSS for styling.
- **Libraries and packages:** FullCalendar for displaying tasks on a calendar, React DatePicker for selecting dates and times, and Lucide React for button icons.
- **Development tools:** Vite for project setup and running the development server, npm for managing packages, and Git/GitHub for version control.
- **Data storage:** Browser localStorage will store tasks as JSON, including each task's title, type, date, and completion status. This will preserve tasks between visits in the same browser.
- **Data sources:** Users will manually enter assignments, exams, and study sessions. The MVP will not require external APIs.

5. Chunked Build Plan

1. Set up the GitHub repository and React project using Vite. Confirm the starter app runs locally.
2. Build the basic page layout with a heading, task form, and calendar area.
3. Add a form for creating assignments, exams, and study sessions with a title, type, and date. Validate required fields.
4. Connect FullCalendar and display newly created tasks on their selected dates.
5. Add editing and deletion so users can update task details or remove tasks. Confirm the calendar reflects each change.
6. Add a completion toggle and visually distinguish completed tasks.
7. Save and load tasks using localStorage. Verify that task details and completion status remain after refreshing.
8. Test the complete workflow, fix any bugs, and update the README with setup and usage instructions.

Each milestone will correspond to one meaningful Git commit.